MSG No. 54.346 P_acca [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `..2/m'`

No.	position	mapping
1	$\left[\frac{3}{4}, 0, 0\right]$	`[1,4,13,16]`
2	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[2,3,14,15]`
3	$\left[\frac{1}{4}, 0, 0\right]$	`[5,8,9,12]`
4	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[6,7,10,11]`

Table 2: Wyckoff site: `4b`, site symmetry: `..2'/m'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,5,12,16]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,6,11,15]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,7,10,14]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,8,9,13]`

Table 3: Wyckoff site: `4c`, site symmetry: `..2/m'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[1,4,13,16]`
2	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,3,14,15]`
3	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[5,8,9,12]`
4	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[6,7,10,11]`

Table 4: Wyckoff site: `4d`, site symmetry: `..2'/m'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5,12,16]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,6,11,15]`
3	$\left[0, 0, \frac{1}{2}\right]$	`[3,7,10,14]`
4	$\left[\frac{1}{2}, 0, 0\right]$	`[4,8,9,13]`

Table 5: Wyckoff site: `4e`, site symmetry: `2'2'2`

No.	position	mapping
1	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[1,4,10,11]`
2	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	`[2,3,9,12]`
3	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[5,8,14,15]`
4	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	`[6,7,13,16]`

Table 6: Wyckoff site: `4f`, site symmetry: `2'22'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,3,10,12]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[2,4,9,11]`
3	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[5,7,14,16]`
4	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[6,8,13,15]`

Table 7: Wyckoff site: `4g`, site symmetry: `2'2'2`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,4,10,11]`
2	$\left[\frac{1}{4},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[2,3,9,12]`
3	$\left[\frac{1}{4},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[5,8,14,15]`
4	$\left[\frac{3}{4},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[6,7,13,16]`

Table 8: Wyckoff site: `4h`, site symmetry: `2'22'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,3,10,12]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[2,4,9,11]`
3	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[5,7,14,16]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[6,8,13,15]`

Table 9: Wyckoff site: `8i`, site symmetry: `2'..`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,10]`
2	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[2,9]`
3	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[3,12]`
4	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{4}\right]$	`[4,11]`
5	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[5,14]`
6	$\left[\frac{1}{2}-x,\ 0,\ \frac{3}{4}\right]$	`[6,13]`
7	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[7,16]`
8	$\left[x+\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[8,15]`

Table 10: Wyckoff site: `8j`, site symmetry: `2'..`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,10]`
2	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[2,9]`
3	$\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[3,12]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[4,11]`
5	$\left[-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[5,14]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[6,13]`

continued ...

Table 10

No.	position	mapping
7	$\left[x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[7,16]
8	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	[8,15]

Table 11: Wyckoff site: 8k, site symmetry: .2'.

No.	position	mapping
1	$\left[\frac{3}{4},\ y,\ \frac{1}{4}\right]$	[1,11]
2	$\left[\frac{1}{4},\ -y,\ \frac{1}{4}\right]$	[2,12]
3	$\left[\frac{1}{4},\ y,\ \frac{1}{4}\right]$	[3,9]
4	$\left[\frac{3}{4},\ -y,\ \frac{1}{4}\right]$	[4,10]
5	$\left[\frac{1}{4},\ -y,\ \frac{3}{4}\right]$	[5,15]
6	$\left[\frac{3}{4},\ y,\ \frac{3}{4}\right]$	[6,16]
7	$\left[\frac{3}{4},\ -y,\ \frac{3}{4}\right]$	[7,13]
8	$\left[\frac{1}{4},\ y,\ \frac{3}{4}\right]$	[8,14]

Table 12: Wyckoff site: 8l, site symmetry: .2.

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{4}\right]$	[1,3]
2	$\left[\frac{1}{2},\ -y,\ \frac{1}{4}\right]$	[2,4]
3	$\left[0,\ -y,\ \frac{3}{4}\right]$	[5,7]
4	$\left[\frac{1}{2},\ y,\ \frac{3}{4}\right]$	[6,8]
5	$\left[\frac{1}{2},\ y,\ \frac{1}{4}\right]$	[9,11]
6	$\left[0,\ -y,\ \frac{1}{4}\right]$	[10,12]
7	$\left[\frac{1}{2},\ -y,\ \frac{3}{4}\right]$	[13,15]
8	$\left[0,\ y,\ \frac{3}{4}\right]$	[14,16]

Table 13: Wyckoff site: 8m, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{3}{4},\ 0,\ z\right]$	[1,4]
2	$\left[\frac{1}{4},\ 0,\ \frac{1}{2}-z\right]$	[2,3]
3	$\left[\frac{1}{4},\ 0,\ -z\right]$	[5,8]
4	$\left[\frac{3}{4},\ 0,\ z+\frac{1}{2}\right]$	[6,7]
5	$\left[\frac{1}{4},\ 0,\ z\right]$	[9,12]
6	$\left[\frac{3}{4},\ 0,\ \frac{1}{2}-z\right]$	[10,11]
7	$\left[\frac{3}{4},\ 0,\ -z\right]$	[13,16]
8	$\left[\frac{1}{4},\ 0,\ z+\frac{1}{2}\right]$	[14,15]

Table 14: Wyckoff site: 8n, site symmetry: ..2'

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	[1,12]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[2,11]
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[3,10]
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	[4,9]
5	$\left[0,\ \frac{1}{2},\ -z\right]$	[5,16]
6	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[6,15]
7	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[7,14]
8	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	[8,13]

Table 15: Wyckoff site: 8o, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{2},\ z\right]$	[1,4]
2	$\left[\frac{1}{4},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[2,3]
3	$\left[\frac{1}{4},\ \frac{1}{2},\ -z\right]$	[5,8]
4	$\left[\frac{3}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[6,7]
5	$\left[\frac{1}{4},\ \frac{1}{2},\ z\right]$	[9,12]
6	$\left[\frac{3}{4},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[10,11]
7	$\left[\frac{3}{4},\ \frac{1}{2},\ -z\right]$	[13,16]
8	$\left[\frac{1}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[14,15]

Table 16: Wyckoff site: 8p, site symmetry: ..2'

No.	position	mapping
1	$[0,\ 0,\ z]$	[1,12]
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	[2,11]
3	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	[3,10]
4	$\left[\frac{1}{2},\ 0,\ z\right]$	[4,9]
5	$[0,\ 0,\ -z]$	[5,16]
6	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	[6,15]
7	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	[7,14]
8	$\left[\frac{1}{2},\ 0,\ -z\right]$	[8,13]

Table 17: Wyckoff site: 8q, site symmetry: ..m'

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,16]
2	$\left[x+\frac{1}{2},\ -y,\ \frac{1}{2}\right]$	[2,15]
3	$\left[-x,\ y,\ \frac{1}{2}\right]$	[3,14]
4	$\left[\frac{1}{2}-x,\ -y,\ 0\right]$	[4,13]
5	$[-x,\ -y,\ 0]$	[5,12]
6	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}\right]$	[6,11]
7	$\left[x,\ -y,\ \frac{1}{2}\right]$	[7,10]

continued ...

Table 17

No.	position	mapping
8	$\left[x+\frac{1}{2}, y, 0\right]$	`[8,9]`

Table 18: Wyckoff site: `16r`, site symmetry: `1`

No.	position	mapping
1	$[x, y, z]$	`[1]`
2	$\left[x+\frac{1}{2}, -y, \frac{1}{2}-z\right]$	`[2]`
3	$\left[-x, y, \frac{1}{2}-z\right]$	`[3]`
4	$\left[\frac{1}{2}-x, -y, z\right]$	`[4]`
5	$[-x, -y, -z]$	`[5]`
6	$\left[\frac{1}{2}-x, y, z+\frac{1}{2}\right]$	`[6]`
7	$\left[x, -y, z+\frac{1}{2}\right]$	`[7]`
8	$\left[x+\frac{1}{2}, y, -z\right]$	`[8]`
9	$\left[x+\frac{1}{2}, y, z\right]$	`[9]`
10	$\left[x, -y, \frac{1}{2}-z\right]$	`[10]`
11	$\left[\frac{1}{2}-x, y, \frac{1}{2}-z\right]$	`[11]`
12	$[-x, -y, z]$	`[12]`
13	$\left[\frac{1}{2}-x, -y, -z\right]$	`[13]`
14	$\left[-x, y, z+\frac{1}{2}\right]$	`[14]`
15	$\left[x+\frac{1}{2}, -y, z+\frac{1}{2}\right]$	`[15]`
16	$[x, y, -z]$	`[16]`